ALPHA CHOICE INNOVATIVE ACADEMY

(International Secondary School)

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GRADE 8

THIRD TERM

AGRIC SCIENCE

MASTERNOTE

NAME: ____

SYNOPSIS

WEEK	TOPIC
1 & 2	Forest and forest uses
3&4	Fishery
3	Development of Agriculture in Nigeria
4	Employment opportunities in Agriculture
5	Soil composition
6	Physical and chemical composition of soil
7	Mid-term break
8	Soilconservation
9	Soil fertility
10	Development of agriculture in Nigeria
11	Farmmachinery

WEEK 1: FOREST AND FOREST USES

OBJECTIVES: At the end of the lesson I will be able to:

i. *Describe a forest*
ii. *Explain forest resources*
iii. *Highlights the uses of forest resources/ products*

CONTENT

A forest can be described as a place where trees, shrubs and smaller plants such as grass grow together. A forest is also a home for many wild animals.

FOREST

Forest resources / products

1. **Forest animals:** Forest animals usually referred to as wildlife, make a major contribution to the diet of not only a majority of rural population in Nigeria, but also of town dwellers. Wildlife meat helps to make up the protein requirements for our healthy growth and well-being.

In Nigeria, forest animal are popularly known as bush meat and the sale of bush meat contributes significantly to the revenue of the country.

2. **Gum-Arabic:** This is obtained from Acacia nilotica and it is used for tanning, dyeing. It is an important base for the manufacture of office gums, food and beverages, confectionaries and pharmaceuticals.
3. **Grasses:** Domesticated animals can be grazed in the forest provided the numbers are kept to a level which does not inhibit regeneration and the area planted with young leaves are fenced.
4. **Latex:** Latex is the rubber obtained from the rubber tree with biological name *Heveabrasiliensis*.

5. **Straw, chewing stick, wrapping leaves, bamboo:** These are all important forest resources. Chewing sticks prevent tooth decay as it contains anti-decay substance known as ***anticariogenic***.
6. **Fibres:** This is converted to ropes and handbags, sponge and twines. They are also used for making nets and fishing lines. They can also be used in making mats, baskets and hats.
7. **Fruits:** Fruits of high nutritive value are obtained from the forest trees such as cashew, apple, mango, orange, banana, guava etc.
8. **Timber:** Timber is used for house building. It is used for furniture manufacturing, bridge construction, manufacturing of simple tools, general construction etc.
9. **Wood pulp:** Wood is converted into cellulose for making papers and other writing materials.
10. **Fuel and firewood:** These are in high demand in Nigeria. This is because 80% of Nigerians reside in rural area and 95% of this populace living in rural areas use firewood for cooking, to keep them warm during cold weather.

Uses of forest

1. Forest provides some of the major human needs such as food, wood for furniture, medicines, etc.
2. Forest trees increases soil fertility when their leaves and stem falls into the ground and decay in the soil.
3. Forest trees helps to prevent soil erosion by means of their roots which bind the soil together.
4. Forests provide employment and career opportunities for a variety of people including foresters, forest guards as well as wildlife and range managers.
5. Forests are a source of revenue to the government.
6. Forest is used as recreational centres or reserves for people who derive pleasure from searching for and looking at the wild animals.

EXERCISE

i. Describe a forest
ii. Explain forest resources
iii. Highlights the uses of forest resources/ products.

WEEK 2: FOREST AND FOREST USES

OBJECTIVES: At the end of the lesson I will be able to:

i. *Explain the effect of forest on the environment*
ii. *Mention and explain the human activities that affect forest production*

CONTENT

EFFECTS OF FOREST ON ENVIRONMENT

1. **Soil conservation:** Forest affects the soil in the following ways:
   i. The trees protect the soil against water erosion; the foliage reduces the speed of the rain drops and gives more time for the soil to absorb water.
   ii. Trees protect the soil from wind erosion; the foliage reduces the speed of the wind in the dry season and prevents surface erosion which could remove valuable top soil inform of dust storms caused by wind.
   iii. Trees shade the soil and reduce high temperature in the dry season. This tends to break the soil surface and facilitate wind erosion.
2. **Climate amelioration:** Forests and their existence produce a more pleasant climatic condition. The air becomes humid and purer. Trees help to break the force of wind, thereby preventing the undesirable blowing of dust and pollution. Trees prevent free evaporation of water, the effect of direct sunlight on soil surfaces and water courses.
3. **Agricultural crop improvement:** Trees may have negative effect on agricultural plants in the immediate vicinity but an overall positive effect on the farm crops. This is due to indirect actions such as soil conservation, soil improvement and water control. The mechanical wind damage to the crop is avoided and the evaporation in the dry season is reduced so that water is saved for the crops to utilize for their growth.
4. **Conservation and regulation of water supply:** The water supply streams, river and waterworks catchment area are made pure and suitable for drinking and for other human uses by the existence of trees in such areas or banks. Even during the rainy period, when large volumes of water flow into the water courses, the bank and the neighbourhood are protected from flooding. Stream flow is clearer. E.g. *Eleyele waterworks Ibadan*.

5. **Soil improvement:**
   i. Trees sustain nutrient by recycling and increasing soil fertility through humus production. The leaves of trees, their twigs, branches falls to the soil and later decay to form humus a vital organic matter which enriches the soil.
   ii. Trees are necessary means of nutrient recycling because rain water percolations during the wet season deposit soil nutrients at a depth that only tree root are able to reach and recycle.
   iii. Some trees species (particularly legumes) fix nitrogen, thus increasing the nitrogen content of the soil which in turn accelerates the humus process of the soil surface

Effects of human activities on forest and its uses

1. **Deforestation:** Deforestation is the reduction of forest cover due to human and other activities. The timber resources of Nigeria have been drastically reduced due to indiscriminate felling and forest clearance for growing food crops and for the construction of roads, homes and for the production of timbers thereby exposing the soil to erosion and other environmental hazard.
2. **Depletion of wildlife:** The wildlife population in Nigeria is decreasing rapidly because of indiscriminate hunting of the animals that are of economic importance in the forest for food, money and other purposes.
3. **Poaching:** Very often people go secretly into the forests to hunt for wildlife and to cut down valuable trees and other plant. This action is called poaching and is strictly against the wildlife conservation law.
4. **Construction of roads:** This is when government cut down trees for road construction or industries.
5. **Bush burning:** This affects the soil greatly in that it does not destroy the trees in the forest alone but also destroy the wildlife animal and destroy the micro-organism in the soil.

EXERCISE

i. Explain the effect of forest on the environment.
ii. Mention and explain the human activities that affect forest production.

WEEK 3&4: DEVELOPMENT OF AGRICULTURE IN NIGERIA

OBJECTIVES: At the end of the lesson I will be able to:

i. *Explain the meaning of fishery*
ii. *List and explain classes of fish*
iii. *Highlight uses of fish and its products*
iv. *Discuss method of fishing*
v. *Mention and explain method of preserving fish*

CONTENT

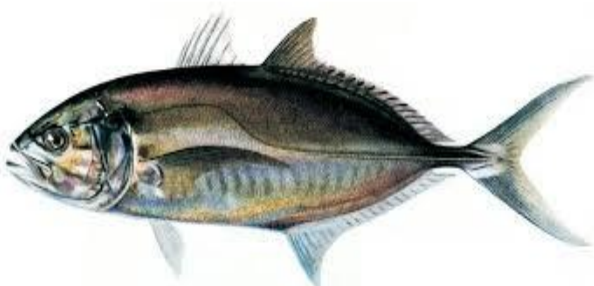
Introduction:

All plants and animals which live in water are describes as aquatic organisms. The branch of fisheries is known as Aquaculture (aqua = water; culture = to grow; therefore, aquaculture means to grow in water).The practice of growing fish in artificial ponds is also called fish farming.

Fishery is the term used to describe the science of catching or growing fish and other aquatic organisms from natural and artificial water bodies.

Fish can be classified based on:

a. **Habitat:** This is the classification based on the environment where the fish lives. It can be freshwater or salt water.
   i. **Freshwater:** This group of fishes live only in fresh water bodies such as rivers, ponds, lakes and swamps. E.g. tilapia, catfish etc.
   ii. **Salt water fish:** This group of fishes are also known as marine fish. E.g. mackerel, shark, croaker, sea catfish, salmon, tuna etc.
b. **Morphology:** This classification based on physical structure of the fish. They are either bony fish or cartilaginous fish.
   i. **Bony fish:** They have stable pattern of bones in their internal skeleton and rooted teeth and operculum which helps them breathe without having to swim. E.g. Catfish, croaker, tilapia etc.



ii. **Cartilaginous fish:** They have a tough skin which is covered with dermal teeth that make the fish feel like sandpaper. E.g. shark, ray, skate, dogfish, etc. other types of aquatic organisms are: shell fish (crabs, crayfish, periwinkles, prawn), Reptile (turtle, crocodile and reptile), mammals (whale, dolphin, turtle).

Uses of fish and its products

1. **Source of food:** Fish and other aquatic organisms are used mainly as human food. Fish flesh is regarded highly for containing first class animal protein, vitamins and mineral salt and other chemical substances that are needed to keep the human body healthy.
2. **Hide and leather:** The skin of cartilaginous fish such as shark is tough and covered with small, sharp spines. It is sometimes dried and specially treated to produce special leather called shagreen. Crocodile and turtle skins also make very good leather for handbags, wallets, belts and shoes.
3. **Polishing materials:** Shells of snails and oysters are used as abrasives for polishing and smoothening surfaces in homes or industries.
4. **Animal feed:** Many fish and parts of fish which are not eaten by humans are processed into fish meal and used in the manufacture of livestock feed. For example, fish meal is an important ingredient of poultry feed used for raising chicken, turkeys and ducks.
5. **Art or fashion ornament:** Clam and oyster shell are used as art ornament for decoration in the homes or public arena, or as earrings, costumes and other jewellery.
6. **Soap and medicines:** The oils obtained from fish, whales and turtles are used as food and also for the manufacture of medicine and soap. Cod-liver oil is a very popular items consumed by many people as a food supplement. Cod liver and other fish oils contain a lot of vitamins.
7. **Glue and fertilizer:** Fish bones are also used for the manufacture of glues and fertilizer.

8. **Building:** Shells of oysters and periwinkles are sometimes mixed with cement and sand for building houses. The periwinkles make the walls stronger and highly attractive.

FISHING TOOLS OR GEARS

Examples of fishing tools or gears commonly used to harvest fish (es) include:

(i) Cast net
(ii) Drag net, draw net or seine net
(iii) Clap net
(iv) Set net
(v) Trawler with seine net or fishing trawler
(vi) Fish traps, gill net, fishing basket, harpoon or spear

FISHING NETS

Fishing nets are generally used for the harvesting of fishes from either ponds, streams, rivers, lakes, oceans or sea. Examples include scoop net, seine net, fishing trawlers, fishing basket; hook and line.

A FISHING NET **HOOK, LINE AND SINKER**

Methods of fishing

a. **Partial harvesting:** This method is done by artisan fishermen. They harvest fish gradually in a desired section of their production area. Fishing tools used are hook and line, baskets, spears, gourd and nets.
b. **Total harvesting:** This method is common in large scale fishing. All fishes and other aquatic animals are harvested at once. It is done through draining of the pond where water is scooped and all fishes harvested.

METHODS OF HARVESTING FISH INCLUDE:

(i) **Trapping:** This involves the setting of traps to catch fish. It involves the use of gears made from ropes or raffia for capturing fishes.

(ii) **Netting:** This involves the use of various types of nets to catch fish. Nets like gill net, clap net etc. are thrown into water to catch fish.

(iii) **Electro-fishing:** This involves the use of electricity to catch fish by creating electric fields in an enclosed area such that current is sent across, killing fishes between poles. It is used only in total harvesting.

(iv) **Draining of ponds:** This involves draining of pond water for easy use of net to catch large fishes.

(v) **Use of ultrasonic:** This is an instrument which can make sound in water and attract fishes so that they can easily be trapped or harvested using other means of harvesting.

(vi) **Impaling:** This involves the use of spears or harpoons to attack and catch big fishes.

Methods of preserving fish

1. **Freezing:** Fish and other aquatic organisms caught are usually preserved by freezing them in ice. This is called ice-fish in Nigeria. Freezing helps to store fish for long period of time, for commercial purpose.
2. **Canning:** Canning is the process of putting fish in tins with special oils or paste. The commonest canned fish are sardines, mackerel and tuna.
3. **Dehydration:** Dehydration is the process of removing moisture with the use of mechanical device that provides artificial heat for drying.
4. **Salting:** Common salt is rubbed on the fish and is also placed inside the fish through the mouth and in the empty belly and dried in the hot sun.
5. **Smoking** The fish to be smoked are placed on metal gauze trays placed on the open top of a mud oven which has a fireplace at the bottom where wood and charcoal are burnt to produce the heat and smoke the fish.

EXERCISE

i. Explain the meaning of fishery.

ii. List and explain classes of fish.

iii. Highlight uses of fish and its products.

iv. Discuss method of fishing.

v. Mention and explain method of preserving fish

WEEK 5: SOIL COMPOSITION

OBJECTIVES: At the end of the lesson I will be able to:

1. *Explain the meaning of soil*
2. *Outline the composition of soil*
3. *Describe the composition by volume*

DEFINITION: Soil can be defined as the topmost part of the earth's crust on which all farming activities are carried out.

TYPES OF SOIL

There are three major types of soil namely;

a. Sandy soil
b. Clayey soil
c. Loamy soil

A. <u>Sandy soil</u>

Soils are referred to as sandy soil if the sand content is predominantly high. It has the following characteristics:

i. It is very gritty/coarse/rough to touch.
ii. It is a loose and single grained particle.
iii. It contains little or no nutrient.

B. **<u>Clayey soil</u>**: Soils are called clay if the clay content is predominantly high. It has the following characteristics;

i. It is smooth when it is touched.
ii. It forms a hard pan when dried.
iii. It is difficult to cultivate.

C. **<u>Loamy soil</u>:** Loamy soil has moderate properties between sandy and clay soil. It has the following characteristics;

i. It contains organic matter.

ii. Not as coarse as sand and not as smooth as clay.

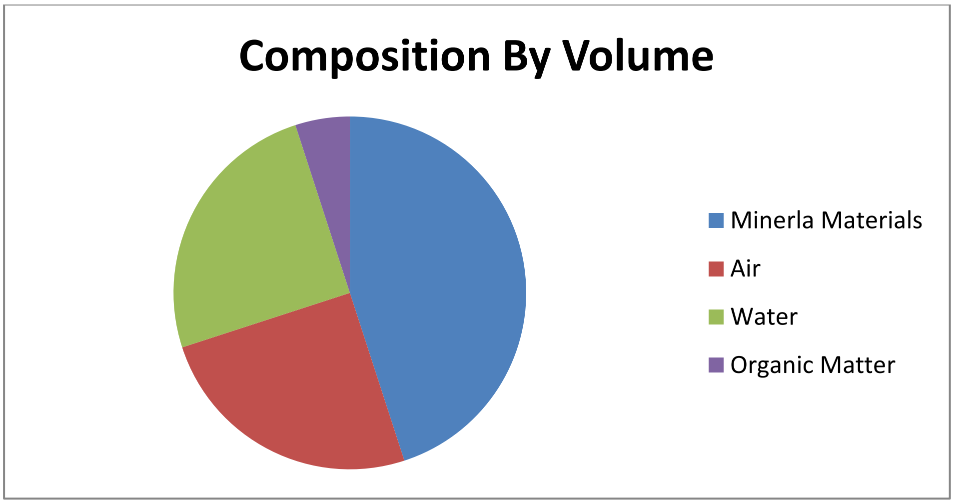
iii. It is good for planting and easy to work on.

COMPOSITION BY VOLUME

The soil consists of the following components in different proportions:

a. Mineral materials (rock materials such as sand, silt, clay and gravel)-45%
b. Organic matter (from dead plants and animals)- 5%
c. Soil water containing dissolved salts- 25%
d. Soil air- 25% and
e. Living organisms such as microorganisms and small soil invertebrates.

The composition of soil by volume can be represented in a diagram below:



EXERCISE

I. Explain the meaning of soil
II. Outline the composition of soil
III. Describe the composition of soil by volume

WEEK 6: PHYSICAL AND CHEMICAL PROPERTIES OF SOIL

OBJECTIVES: At the end of the lesson I will be able to:

i. *Explain the meaning of physical properties of soil*
ii. *Highlight the physical properties of soil*
iii. *Discuss soil structure, profile, texture and soil consistency*
iv. *Enumerate the chemical properties of soil*
v. *Explain the meaning of soil acidity and alkalinity*
vi. *Describe the effects of soil acidity and how it can be corrected*

CONTENT

The physical properties of soil are properties that can be easily evaluated by visual inspection or by feeling. They can easily be measured by some kind of scales such as size, shape and strength or intensity. They include

1. **Soil texture:** This is the degree of fineness or coarseness of the mineral particles of a soil. This can also be defined as the relative proportions of sand, silt and clay in the soil.

Importance/effects of soil texture

i. Determines the capacity of the soil to hold water and to circulate air.

ii. Determines the amount of surface area available for reaction.

iii. Determines soil temperature

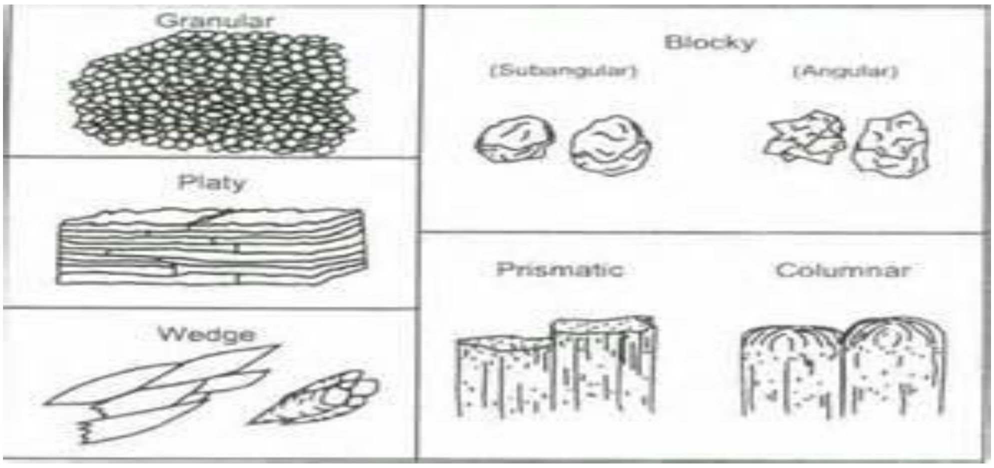
2. **Soil structure:** This is the physical appearance of the soil according to the arrangement of the individual particles. In simple terms, it is the way in which sand, silt, and clay group themselves to form aggregates or clusters.

The types of soil structure include:

i. Plate like structure
ii. Prismatic Structure
iii. Block like structure
iv. Columnar structure

v. Crumb structure

vi. Granular structure



Diagrams Showing Different Soil Structure Types

Importance/effects of soil structure in agriculture

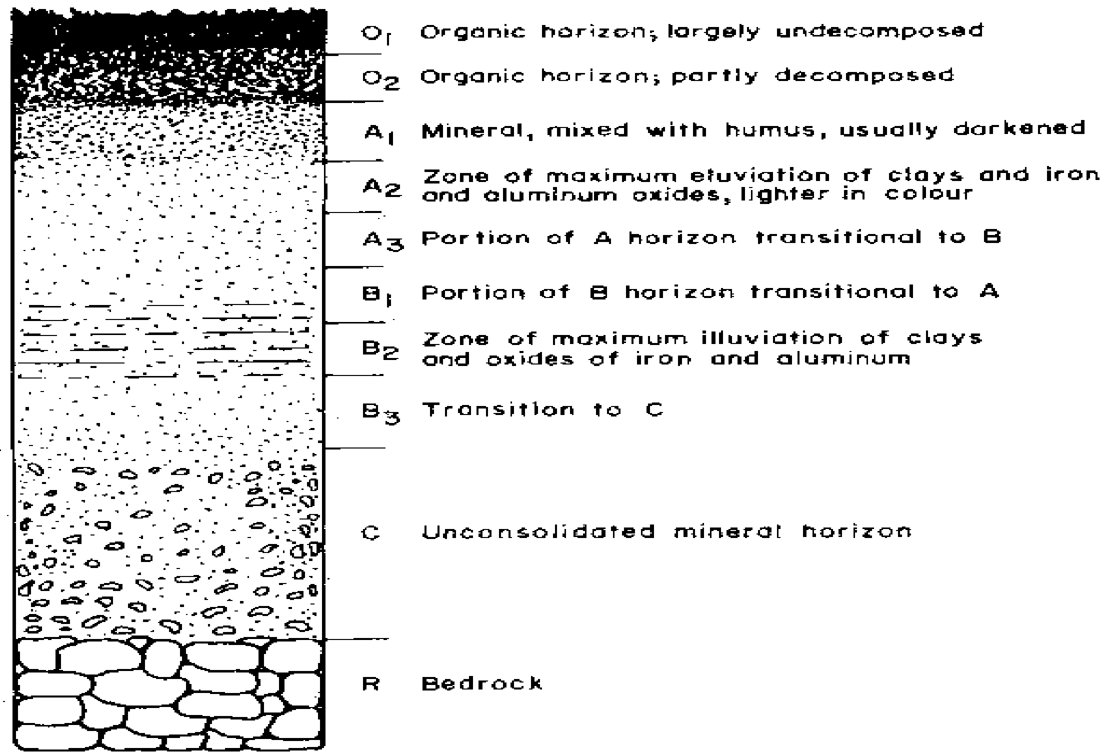
i. Increases the aeration of the soil and water holding capacity of the soil.
ii. Facilitates the activities of micro-organisms.
iii. It reduces erosion by allowing optimum water percolation and thus preventing water-logging.

3. **Soil consistency:** This refers to the ability of the soil to hold/stick together. The degree of cohesion of a soil, its resistance to deformation when worked on or its resistance to the insertion of an instrument when worked on or its resistance to the insertion of an instrument like a cutlass, shovel, knife or hoe is a measure of its consistence.

Important Effects of Soil Consistency

It is useful in estimating the support strength of soils under applied force

4. **<u>Soil profile</u>:** A collection of vertical section of the soil showing different layers of soil called horizons.



Soil Profile Diagram

The chemical properties of soil include:

Soil pH (*Pondus de Hydronium*): This is a measure of acidity or alkalinity of a soil.

Soil acidity: This is the concentration of hydrogen ions in the soil. **Soil alkalinity**: This refers to the concentration of hydroxyl ions in the soil.

1 2 3 4 5 6 7 8 9 10 11 12 13 14

Acidity increases Neutral Alkalinity increases

A pH scale

<u>Causes of soil acidity</u>

i. Application of nitrogen (acid)- containing fertilizers such as Ammonium sulphate and Ammonium nitrate

ii. Nutrient up-take by plants and leaching of soil

iii. Presence of sulphur in the soil

Figure 1. Soil pH and Plant Growth

Soil Reaction	pH	Plant Growth
	>8.3	Too alkaline for most plants
	7.5	Iron availability becomes a problem on alkaline soils.
Alkaline soil	**7.2**	6.8 to 7.2 – "near neutral"
Neutral soil	**7.0**	6.0 to 7.5 – acceptable for most plants
Acid soil	**6.8**	
	6.0	
	5.5	Reduced soil microbial activity
	<4.6	Too acid for most plants

<u>Effects of soil acidity</u>

1. It reduces micro-organism and their activities
2. It causes nutrient deficiency in the soil e.g. Ca, P (calcium and phosphorous)
3. Nutrient can be leached out of the soil which results in reduced growth and yield of crops

<u>Correction/Removal of soil acidity</u>

1. Soil acidity can be corrected or controlled by the application of lime materials such as:Limestone- $CaCO_3$, Slaked Lime- $Ca(OH)_2$, Quick Lime- CaO, Calcium Hydrogen Trioxocarbonate (IV)- $Ca(HCO_3)_2$, Wood ash- Calcium Phospholicate, Basic Slag.
2. Farmers should not apply excess acidic fertilizers to the soil.

Soil acidity can be determined by using:

1. Litmus paper
2. BDH universal soil indicator method
3. Colorimetric Method and

4. Electrometric method

EXERCISE

i. Enumerate the chemical properties of soil
ii. Explain the meaning of soil acidity and alkalinity
iii. Describe the effects of soil acidity and how it can be correct

WEEK 8: SOIL CONSERVATION

OBJECTIVES:

1. Explain soil conservation.
2. State importance of conserving soil.
3. Mention and explain methods of conserving soil.
4. *State factors responsible for the loss of nutrients.*
5. *Identify some human activities which contribute to loss of nutrient.*
6. *Highlight effect of soil conservation.*

7.

Soil conservation is the intelligent use of soil so that enough plant nutrients are available to plants for growth and production at all times.

The importance of soil conservation includes:

(i) Protection of the environment from destruction.
(ii) Prevention of soil erosion
(iii) Supporting wild life
(iv)Provision of space for erection of building
(v) Micro organisms and mineral resources are found in the soil.
(vi) it reduces soil acidity
(vii) it improves water infiltration in the soil and reduces runoffs
(viii) it improves the soil structure
(ix) it reduces leaching and contamination of ground water.

METHODS OF SOIL CONSERVATION

1. **Crop rotation:** This is the system of planting different types of crops on the same piece of land such that they follow one another in definite order. The rotational cycle of the crop is planned in such a way that the nutrients that are removed from the soil are replaced by another crop planted afterwards.

   An example of four years rotation is as follows:

Plot	2013	2014	2015	2016
1	Maize	Groundnut	Yam	Cowpea
2	Groundnut	Yam	Cowpea	Maize
3	Yam	Cowpea	Maize	Groundnut
4	Cowpea	Maize	Groundnut	Yam

2. **Application of organic manure:** Organic manure includes farmyard manure, poultry manure, compost manure and green manure. It adds or replaces lost soil nutrients. It reduces the risk of soil erosion and leaching due to its binding effect.

3. **Application of inorganic manure:** inorganic manure is fertilizers, which are chemically synthesized in the form of powders, pellets or crystals to correct certain deficiencies in the soil.

4. **Bush fallowing:** it involves the process of leaving a piece of land to rest and regain its fertility after a thorough use. The fallow period could be 6-12 years before it is used again.

5. **Planting of cover crops:** This is the process by which certain crops that are leguminous in nature are planted to provide a protective cover over the soil. Examples of such crops are melon, cowpea, mucuna.

6. **Mulching:** Mulching is a method used in preventing evaporation or evapo-transpiration by the placement of straw, leaves, saw-dust e.t.c. on the surface of the soil. The dry leaves also enrich the soil fertility and reduce erosion and rain splash.

7. **Liming:** It is the addition of lime to the soil. This is purposely done to increase the pH value of the soil especially if the soil is acidic in nature.

Factors responsible for the loss of soil nutrients

Soil pH: This is the degree of acidity or alkalinity of the soil. High soil pH leads to accumulation of calcium and magnesium ions which affect crop growth.

Erosion: It completely removes the top soil which contains humus and reduces the population of soil living organisms. It also destroys soil structure.

Leaching: Leaching is the washing down of soil nutrients to region below the root zone of the plant.

Flooding: This is a situation whereby a large amount of water covers a dry farmland. Flood causes damage to crops by retarding their growth.

Soil pH: This is the degree of acidity or alkalinity of the soil. High soil pH leads to accumulation of calcium and magnesium ions which affect crop growth.

Human activities responsible for loss of nutrients

Soil pH: This is the degree of acidity or alkalinity of the soil. High soil pH leads to accumulation of calcium and magnesium ions which affect crop growth.

Crop removal (clean clearing): Harvesting crops without incorporating the crop residues into the soil leads to depletion of soil nutrients.

Bush burning: It destroys microbes and organic matter contents of the soil. It also exposes the soil to erosion and leaching.

Building of farm structures: Soil is destroyed when the foundations of structures are being layed on the land. In the process of gathering soil for brick making, the soil surface is exposed to intensive heat and cover crops and mulching materials are removed. All these affect soil adversely.

Deforestation: This is the continuous felling of forest trees without replacing them. It increases soil erosion.

Continuous cropping: Continuous uptake of nutrients by crops without fallow periods can lead to the depletion of soil nutrients.

Fertilizer application: Excessive application of fertilizers can increase soil acidity, reduces the activities of soil organisms and hinders crop growth and development.

Erosion: It completely removes the top soil which contains humus and reduces the population of soil living organisms. It also destroys soil structure.

Effects of soil conservation practices on the environment

Pollution of ground water: Excessive application of organic manure of inorganic fertilizers to the soil can contaminate ground water by leaching.

Prevalence of aquatic weeds: The washing away of fertilizers from the farm lands to the streams and rivers leads to the growth of aquatic weeds.

Spread of diseases: Toxic liming materials can be washed into streams and rivers. These polluted water bodies are not fit for animal and human consumption.

EXERCISE:

1. Define soil conservation.
2. Mention and explain five methods of soil conservation.
3. Highlight six importance of soil conservation.
4. State factors responsible for the loss of nutrients
5. Identify some human activities which contribute to loss of nutrient
6. Highlight effect of soil conservation

WEEK 9: SOIL FERTILITY

OBJECTIVES; At the end of the lesson, I will be able to;

1. *Define soil fertility*
2. *State and briefly explain the factors that affect the availability of soil nutrients*
3. *Mention the ways of maintaining soil fertility*
4. *Differentiate between micro and macro nutrients*

INTRODUCTION: Soil is the main principal source of plan nutrients and water. Crops absorb dissolved nutrients with their roots which are firmly anchored into the soil. However, continuous cultivation of crop without soil management absorbs the soil nutrients and reduces crop yield.

SOIL FERTILITY

Soil fertility: refers to the ability of the soil to supply essential plant nutrients and soil water in adequate amounts and proportions for plant growth and reproduction. A soil is said to be fertile when it has all the required plant nutrients and microorganisms and is free of toxic materials.

Factors that affect the availability of plant nutrient in the soil

1. **Soil pH**: This is the degree of acidity or alkalinity of the soil. High soil pH leads to accumulation of calcium and magnesium ions which affect crop growth.
2. **Leaching:** This is the removal of nutrients from the top soil to the subsoil by percolating water beyond the reach of plant roots. It leads to accumulation of hydrogen and aluminium ions which are acidic and toxic to plants.
3. **Crop removal**: Harvesting crops without incorporating the crop residues into the soil leads to depletion of soil nutrients.
4. **Bush burning**: It destroys microbes and organic matter contents of the soil. It also exposes the soil to erosion and leaching.
5. **Erosion**: It completely removes the top soil which contains humus and reduces the
   population of soil living organisms. It also destroys soil structure.

Ways of maintaining soil fertility

1. **Bush fallowing**: this is a system of farming in which a land that has been cultivated for years is allowed to rest for three to five years. During this period, the land would have regained its fertility.
2. **Planting of cover crops:** Cover crops are crops that protect soil against erosion. The crops that are used are mainly legumes. The dead leaves of these plants when decomposed increase soil fertility and create conducive environment for increase microbial activities in the soil.
3. **Liming:** This is the application of calcium and magnesium compounds to the soil to reduce soil acidity.
4. **Mixed cropping:** A leguminous crop such as cowpea, soya beans add nitrogen to the soil and improves soil fertility. Crops with different rooting systems will extract nutrients from different depths of soil.

5. **Mulching:** In this case, the surface soil is covered with straw or any other cut vegetative material. It improves soil structure, conserve soil moisture and regulate soil temperature
6. **Organic fertilizer:** this refers to any nutrient containing materials derived from either plant or animal origin or both e.g. compost manure, farm yard manure e.t.c.
7. **Inorganic manure:** This refers to the chemical fertilizers which are manufactured industrially.

Characteristics of a fertile soil

1. It is rich in nutrients necessary for basic plant nourishment. This includes nitrogen, phosphorus and potassium.
2. It consists of adequate minerals such as boron, chlorine, copper, iron, magnesium, sulphur, manganese, cobalt, zinc. These minerals promote plants nutrition.
3. It contains soil organic matter that improves the structure of the soil. This enables the soil to retain more moisture.
4. The soil **pH** is in the range 6.0 – 6.8.
5. It has a good soil structure which result in well-drained soil.
6. It consists of a variety of micro-organism that support plant growth.

Types of plant nutrient

There are two main types of plant nutrients:

1. **Macro-nutrients (essential elements)**: These are nutrients required by crops in large quantities for proper growth and development. Examples are potassium, calcium, sulphur, nitrogen, phosphorus, e.t.c.
2. **Micro-nutrients (non-essential elements):** these are nutrients required by plants in small quantities for their healthy growth. Examples are manganese, zinc, chlorine, copper, boron, iron, e. t. c.

EXERCISES;

1. What is soil fertility?
2. What the factors that affect the availability of soil nutrients?
3. What are the ways of maintaining soil fertility?
4. State the differences between micro and macro nutrients

WEEK10:DEVELOPMENT OF AGRICULTURE IN NIGERIA

OBJECTIVES: At the end of the lesson I will be able to:

i. *List and explain the Roles of Science and Technology in the Development of Agriculture in Nigeria.*
ii. *Factors affecting the application of science and technology in agriculture.*

CONTENT

INTRODUCTION

Any country that desires to develop its agricultural system must among other things apply science and technology in the production, processing, preservation and marketing of food and fibre.

Science can be defined as knowledge obtained through the observation and testing of fact, which is arranged in an orderly manner.

Technology is the application of scientific knowledge for the production of useful things.

Roles of Science and Technology in the Development of Agriculture in Nigeria.

1. **Farm power and machinery:** This is also known as mechanization. This has led countries into believing that they can solve their food problems through the mechanization of agriculture. Farmers were given loans by the government to purchase agricultural machinery and equipments.

MILKING MACHINE **HARVESTER**

2. **Improved planting materials (e.g. seeds and seedlings):** Through the help of science and technology, improved planting materials have been developed. Improved seeds and seedlings are being developed by scientists to boost agricultural production. These seeds and seedlings do not only mature early but also are high yielding and disease resistant.

3. **Biotechnology:** We can describe biotechnology as the modification of living things, that is, plant and animals, to make useful and improved products that can be beneficial to human beings, production of vaccines etc. For example, they can produce better plants that give high yields, production of improved livestock in animal production. There are three major areas of biotechnology;
    i. **Tissue culture:** This is the technique of growing whole new plant from very tiny parts (tissue) of plants in a laboratory in specially formulated media containing nutrients. When the young plants (**plantlets**) mature in the laboratory, they are transferred to the farm where they grow normally to give high yields.
    ii. **Plant genetic modification:** The research scientists add some new genes into the plant to improve them so that the pests or diseases do not like them anymore. In this way, the improved cowpea, plantain or cassava plant can grow better and give high yields.
    iii. **Molecular breeding:** This is when scientists identify and choose certain plants based on their very good and useful properties such as high yields and ability to grow well. These good properties are called ***traits*.** So when scientists do this, they can produce crops with good properties or even remove traits that are not good from crops.
4. **Production of chemical fertilizers:** The development and application of chemical and organic fertilizers is a direct result of application of science and technology. These chemical increase soil fertility and so the land does not have to be left to fallow for many years and can support crop production for many years.
5. **Pest and disease management:** Scientists have now produced many types of chemicals for the control of crop and animal pests, diseases and weeds. Pest control materials have also been produced from plants. Such plant products are called **Biopesticides.** They are safer to use than chemicals pesticides.

6. **Harvesting, processing and packaging:** In many large farms now, crops are harvested by the use of machines as a result of scientific and technological inventions.
   Through the application of science and technology, food processing has been made easy so women can now buy yam flour in plastic bags to make pounded yam, or plantain flour, semolina or amala flour to prepare food easily. These processed food items can be well stored for long periods without spoiling. Example milk from cows, is now well preserved as fresh milk or powdered milk. Fruit juices are also now available in cartoons which we buy every day. Foods are now canned.

7. **Storage of agricultural produce:** If the storage structures and methods are not good enough, pest and diseases and even the weather can cause a lot of damage to the harvested crops in storage. Through the application of science and technology, scientists have developed very efficient storage system for different crops.

INCUBATOR

8. **Information and communication technology:** ICT involves acquiring agricultural information by using computers, mobile or cellular phones, listening to agricultural programmes on radio, watching programmes on television or video showing the best way of growing crops, taking care of livestock or even selling agricultural produce.
9. **Transportation:** Science and technology helps in the construction of transportations networks such as roads, railways and canals to link one village to another and help the farmers transport farm produce to cities.
10. **Water supply:** With the help of science and technology, bore holes, deep wells and water reservoir have been constructed to provide steady water supply of water to farms and villages.

Factors affecting the application of science and technology in agriculture

1. **Availability of funds:** Farm machines and tools are expensive and usually not within easy reach of small holder farm. Mechanised farming also required plenty fund.
2. **Availability of farming land:** In most Nigerian communities, land is a limiting factor in agricultural production. Most land in Nigeria is fragmented making mechanised agriculture difficult.
3. **Slow willingness by farmers to adopt scientific methods:** For science and technology to be fully applied, farmers must be willing to adopt the improved methods of farming. Most of the traditional farmers are conservatives and are very reluctant to adopt new scientific farming methods and techniques because they are afraid to take risks with new technologies.
4. **Education and training:** For farmers to adopt modern methods of farming, they need to be educated to prepare their minds for the adoption and application of scientific and technological innovations in farming. If farmers are educated, extension agents find it easier to train them in improved agricultural production practices, storage, processing and marketing of other produce.
5. **Availability of market information and facilities:** The proper development and use of Information and Communication Technologies (ICT) to disseminate agricultural information can help Nigerian farmers to increase their food and agricultural production as well as improve their marketing technique.

EXERCISE

i. List and explain the Roles of Science and Technology in the Development of Agriculture in Nigeria.

ii. Factors affecting the application of science and technology in agriculture.

WEEK 11: FARM MACHINERY

OBJECTIVES: At the end of the lesson I will be able to:

1. *Define farm machinery*
2. *Explain the meaning of tractor coupled implement*
3. *Discuss types of farm machines and tractor coupled implement*

CONTENT

FARM MACHINERY

These are various types of machines and implements used on the farm to make difficult jobs easy to carry.

TRACTOR COUPLED IMPLEMENTS

These are implements that are attached to the tractors to perform farm operation. E.g. plough, harrows, ridgers, planter, sprayer, harvester, etc.

TYPES OF FARM MACHINES BASED ON USES

i. **Primary machine:** These are machines which supply mechanical power to other machines for farm operation e.g. tractors.
ii. **Tillage machines:** They are used for the preparation of the soil before planting. They include ploughs, harrows, ridgers.
iii. **Planting machines:** These are machines used to plant or sow planting materials (seeds, stems etc.) in the soil. Examples are broadcasters, drillers and planter.

Iv Processing machine: These are machines such as grinders, hammer mills, shellers, etc.

i. **Milking machines:** These are machines used for extracting milk from the udders of dairy farm animals such as cow, doe and ewe.


ii. **Incubators:** These machines are used for keeping fertile eggs under controlled temperature until hatching.


iii. **Harvesting machines:** These are the machines used to remove the mature parts of crops. E.g. corn pickers, mowers, combined harvesters.


Types of Tractor coupled implements

1. **Ploughs:** They are primary tillage implement used for loosening the soil before seeds are sown. There are different types of ploughs

i. **Disc plough:** It is made up of concave disc of diameters ranging between 50- 80cm. the disc cut, elevates and turn the soil. Disc plough is suited for tropical soil than mould-board plough. The disc roll over stumps root and rocks.

ii. **Mould-board plough:** This is made up of frame, mould- board and standard share. It is adapted for use in temperate or light soil where there are no hard rocks or roots.

2. **Ridgers:** They are used for making ridges. Some common examples of ridgers are: disc ridgers, mould- board ridgers.
3. **Planters:** These are machines that are used to plant seeds, tubers and seedlings and stem cuttings in the soil. There are different types of planters. Some common examples are: broadcasters, row crop planters, precision planters and grain drill planters.
4. **Harrows:** They are secondary tillage implements used for harrowing. They are used soil clods into fine textures for planting seeds. There are different types of harrows;
   a. **Disc harrows**: They consist of sets of concave discs of different diameters. The edges of the discs can be plain or serrated.
   b. **Spike-tooth harrow**: This has several metal pegs attached to a mesh framework.
   c. **Spring-tine harrow**: This has C-shape tines arranged in two or three rows.
5. **Sprayers**: These are equipment designed for spraying agro-chemicals to control pests, diseases and weeds. The different types of sprayer are knapsack sprayer, tractor mounted (boom) sprayer, helicopter mounted sprayer.

EXERCISE

i. Define farm machinery

ii. Explain the meaning of tractor coupled implement

iii. Discuss types of farm machines and tractor coupled implement

Maintenance of Farm Machines

Maintenance involves the different operations performed on machines, implements, devices, and tools to prolong their useful life and to keep them in good working conditions. Maintenance can be classified as preventive or corrective maintenance. Corrective maintenance is costlier than preventive maintenance because it involves the cost of buying new parts and replacements. Farm machines can be maintained in the following ways:

i. Read user manuals before operating any machine
ii. Use the implements or machines for the right operations.
iii. Wash and dry farm implements after use
iv. Tighten loose bolts and nuts and replace damaged parts with new ones.
v. Lubricate movable part of farm machines to prevent wear and tear.
vi. Store farm machines and implements under sheds in cool or dry place
vii. Paint or grease metallic parts before storage.

Uses of farm machines

1. They contribute to the quantity and quality of farm produce
2. They allow increase in the size of the farm produce
3. They can be used with trailers to carry harvested crops or other products from the farm to the market
4. They are coupled with trailers to transport workers within the farm
5. They can be used to control weeds and bury plant residues
6. They help to mix the soil together
7. They improve aeration and water infiltration of the soil
8. They cut and turn the top soil and improve weathering
9. They help to control erosion

EXERCISE

1. Define farm machinery
2. Explain the meaning of tractor coupled implement
3. Discuss types of farm machines and tractor coupled implement